

Machine Learning and Pattern Classification

Challenge:
Team Observe



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Evaluation setup

- Dataset split:
 - 8230 files total; 60% train; 20% val; 20% test
- Baseline classifier:
 - Average cost: 108.8553

Model Architecture

- Model Class: TCN (Temporal Convolutional Network)
 - More efficient than RNN for long-term dependencies.
 - Doesn't rely on sequential processing (Faster training and easier to parallelize).
 - Less likely to overfit.
- Loss Function: Weighted BCE-Loss

Layer	Input Shape	Output Shape	Details
Convolutional Feature Extractor			
Input	$[B, F, D]$	$[B, F, D]$	Raw frame-level features
Transpose	$[B, D, F]$	$[B, D, F]$	Prepare for Conv1D over embedding axis
Conv1D + BN + ReLU	$[B, D, F]$	$[B, C, F]$	1 st Conv1D block, kernel size = 3, padding = 1
Dropout	$[B, C, F]$	$[B, C, F]$	Dropout rate = 0.2
Conv1D + BN + ReLU	$[B, C, F]$	$[B, C, F]$	2 nd Conv1D block, kernel size = 3, padding = 1
TCN (Temporal Convolutional Network)			
ResidualBlock ($\times$ num_blocks)	$[B, C, T]$	$[B, C, T]$	Each block: $2 \times$ Conv1D + BN + ReLU + Dropout = 0.2, dilations = 2^i
Conv1D (output layer)	$[B, C, T]$	$[B, N, T]$	Kernel size = 1
Transpose	$[B, N, T]$	$[B, T, N]$	Final output for framewise classification

Strategies to improve performance: Variable Thresholds

- Globally constant threshold ignores class imbalances.
- Threshold search space consists of values from 0 to 1.
- Thresholds are tuned after each validation epoch.
 - Utility function: $U = TP - cost_{FP}FP - cost_{FN}FN$

Strategy	Best Validation Cost
baseline (no strategy)	40.1256
Variable Thresholds	38.275
Post-processing	40.5103
Tuned Parameters	39.8219

Strategies to improve performance:

Post Processing

- Spurious positive class predictions lasting only a few frames could have a big effect on overall performance.
 - Amplified by weighted loss function: Model more pushed toward predicting positive.
- Post-processing strategy:
 - Predictions lasting only a few frames get discarded.
 - Best result for rejecting positive predictions lasting less than 3 frames.

Strategy	Best Validation Cost
baseline (no strategy)	40.1256
Variable Thresholds	38.275
Post-processing	40.5103
Tuned Parameters	39.8219

Strategies to improve performance: Hyperparameter Tuning

- Tested hyperparameters:
 - Optimizers: Adam, AdamW
 - Batch sizes: 32, 64, 128, 256
 - Learning rate: 1e-3, 1e-4, 1e-5
- Best hyperparameter combination: AdamW, 64, 1e-4

Strategy	Best Validation Cost
baseline (no strategy)	40.1256
Variable Thresholds	38.275
Post-processing	40.5103
Tuned Parameters	39.8219

Strategies to improve performance: Improvement on cost

- A combination of the strategies was used

